```
# chown -R mirror:mirror /var/www/html/pub
# find /var/www/html/pub -type d -exec chmod 0755 \{\} \;
```

If you wish to exclude some content from synchronising, you will create an **exclude.txt** file. You may put any expression into that file and when **rsync** is told about it, it won't pull that content. You can do this as your new *mirror* user:

```
# su - mirror
$ touch exclude.txt
```

An **exclude.txt** file typically looks like what follows:

```
#dont sync any ppc content
ppc*

#don't sync debug directories
debug*

#don't sync source directories
source*
```

As you can see, you can put regular expressions in the **exclude** file. It means that you need not put in all the names of the directories that you want to exclude. When you put **ppc*** in the **exclude.txt** file, all directories starting with ppc will not be pulled.

Now that we are finished with the **exclude** part, we are ready to pull in the actual content. The **rsync** command may look like what's given below:

```
$ rsync -vaH —exclude-from=/home/mirror/exclude.txt \
 —numeric-ids —delete —delete-after –delay-updates \
 rsync://mirror.anl.gov/fedora/linux/releases/11 \
 /var/www/html/pub/fedora/linux/releases/
```

This command will start pulling the Fedora 11 repository and put them into **/var/www/html/pub/fedora/linux/releases/11**.

Now, let's see what this means. **rsync**, as stated earlier, is an incremental file transfer protocol. **-v** stands for verbose mode, **-a** means the **achieve** option, and **-H** means that the **rsync** run will preserve hard links between the files (which saves considerable amounts of disk space and reduces file transfers).

We now define which directories not to synchronise using **—exclude-from**. The **—delete**, **—delete-after** and **—delay update** tells **rsync** not to delete old content while synchronising new data. Instead, it tells **rsync** to keep the old file and directories until the synchronisation is complete. Then, finally, we define the remote **rsync** server and the destination directory.

If you are worried from which server you want to pull the repositories from, you can get a list of servers, which provide the **rsync** service, from the Fedora mirrorlist at **http://mirrors.fedoraproject.org/publiclist/**. It would be nice to choose a reliable server near you. Also, don't forget to drop a mail to the admin of the server, as a matter of courtesy and also to ensure there is no planned outage in the next couple of days, at their end.

Saving some bandwidth

A little trick can save you a few gigabytes of download. If you are not sure about the directory structure Fedora repositories have, be a bit careful about this.

The ISO of the Fedora DVD resides at the **Fedora/$architecture/iso/** directory. Also, the same contents of the DVD are at **Fedora/$architecture/os/**, but as extracted files and directories. For example, **http://118.102.181.66/releases/11/Fedora/i386/os/** contains the files of **http://118.102.181.66/releases/11/Fedora/i386/iso/Fedora-11-i386-DVD.iso**. So if you download the ISO image first and then copy the content over to the **os/** directory, you need not download the same content twice. Let's see how we do it.

Once the download of the DVD ISO file is completed, mount it somewhere:

```
# mount -o loop /var/www/html/pub/fedora/linux/releases/11/Fedora/i386/
iso/Fedora-11-i386-DVD.iso /mnt
# cp -prv /mnt/* /var/www/html/pub/fedora/linux/releases/11/Fedora/i386/
os/
# umount /mnt
```

Similarly, you can repeat this for x86_64 DVD ISO, if you are mirroring that architecture too.

> **Note:** Be sure you use the -p option with *cp*. If you don't, the copy operation will change the timestamps of the files being copied and *rsync* will treat them as invalid. *rsync* will pull all the content again, overwriting the copied files, and in the process thwart all your efforts to save bandwidth.

If the download stops

In the course of synchronising, it is highly possible that you will receive a few messages like this: "**Suddenly the Dungeon collapses!! - You die...**" and the download will stop. Don't panic. It's only that **rsync** has stopped for some reason. Just press the up arrow key and press **Enter** to run the same command again. **rsync** will pick up from where it left off. Also, you won't be able to see any file in the directories until all the content of a directory is pulled. You can be assured that the download is indeed happening by using this feature periodically:

```
# du -m /var/www/html/ | tail -n 1
```

Let **rsync** run its own course. You have nothing to do other than periodically check if it has stopped. In the meantime, let's do the other necessary configurations.